Chapter 2: Financial Statements, Taxes and Cash Flow

Multiple Choice

1. A firm has net sales of $2,400, cost of goods sold of $1,300, depreciation of $200 and interest paid of $150. The tax rate is 21%. What is net income?
a. $900.00
b. $711.00
c. $592.50
d. $750.00

2. A Korean firm has taxable income of ₩500 million. Using the national corporate tax schedule in the table, what are its average and marginal tax rates?

Taxable income	Rate
Up to ₩200 million	10%
₩200 million to ₩20 billion	20%
₩20 billion to ₩300 billion	22%
Over ₩300 billion	25%

a. Average 15%; marginal 20%
b. Average 16%; marginal 20%
c. Average 16%; marginal 10%
d. Average 20%; marginal 20%

3. A firm issues 2,000 shares of common stock with a par value of $0.50 at an issue price of $12 per share. How much goes to additional paid-in capital (paid-in surplus)?
a. $1,000
b. $24,000
c. $22,000
d. $23,000

4. Zephyr Logistics, 2025: EBIT $2,600, depreciation $1,000, taxes $420. Net fixed assets rose from $9,000 to $9,400. Current assets rose from $5,000 to $5,800 and current liabilities rose from $2,500 to $2,700. What is the cash flow from assets (CFFA)?
a. $1,180
b. $1,780
c. $2,380
d. $3,180

5. A firm paid interest of $100 and dividends of $885. Its long-term debt rose from $3,000 to $3,200 and its common stock and paid-in surplus rose from $2,500 to $2,600 because of a new share issue. What are its cash flow to creditors and cash flow to stockholders?

a. −$100 and $785

b. $300 and $985

c. $100 and $885

d. $785 and −$100

Chapter 2: Financial Statements, Taxes and Cash Flow — Answers

1. c
EBIT = 2,400 − 1,300 − 200 = 900; taxable income = 900 − 150 = 750; taxes = 21% × 750 = 157.50; net income = 750 − 157.50 = $592.50.

2. b
Tax = 10% × 200M + 20% × 300M = ₩80M, so the average rate = 80 / 500 = 16%; the marginal rate is the 20% bracket that applies to the next won of income.

3. d
Paid-in surplus = (issue price − par) × shares = (12 − 0.50) × 2,000 = $23,000; the $1,000 of par goes to the common stock line.

4. a
OCF = 2,600 + 1,000 − 420 = 3,180; NCS = 9,400 − 9,000 + 1,000 = 1,400; ΔNWC = (5,800 − 2,700) − (5,000 − 2,500) = 600; CFFA = OCF − NCS − ΔNWC = 3,180 − 1,400 − 600 = $1,180.

5. a
CF to creditors = interest paid − net new borrowing = 100 − 200 = −$100; CF to stockholders = dividends paid − net new equity raised = 885 − 100 = $785; their sum, $685, equals CFFA.

Chapter 3: Working with Financial Statements

Multiple Choice

1. Coastal Furniture has inventory of $84,600, equity of $351,000, total assets of $470,000 and sales of $612,300. What is the common-size percentage for inventory?
a. 74.7%
b. 24.1%
c. 18.0%
d. 13.8%

2. A firm has cash of $98, current assets of $708 (including inventory of $422) and current liabilities of $548. What is its quick ratio?
a. 0.18
b. 0.52
c. 1.29
d. 0.77

3. A firm has total assets of $3,630 and total equity of $2,625. What are its debt/equity ratio and its equity multiplier?
a. D/E = 1.38; EM = 0.38
b. D/E = 0.28; EM = 1.28
c. D/E = 0.28; EM = 0.38
d. D/E = 0.38; EM = 1.38

4. A firm has EBIT of $741, depreciation of $276 and interest of $141. What are its times interest earned (TIE) and cash coverage ratios?
a. TIE = 3.30; cash coverage = 5.26
b. TIE = 7.21; cash coverage = 5.26
c. TIE = 5.26; cash coverage = 3.30
d. TIE = 5.26; cash coverage = 7.21

5. A firm has sales of $2,361, cost of goods sold of $1,344, inventory of $422 and accounts receivable of $188. What is its inventory turnover?
a. 3.18
b. 0.31
c. 7.15
d. 5.59

6. A firm has net income of $474, sales of $2,361, total assets of $3,630 and total equity of $2,625. What are its profit margin, ROA and ROE?
a. 18.06%, 13.06%, 20.08%
b. 13.06%, 20.08%, 18.06%
c. 20.08%, 13.06%, 18.06%
d. 20.08%, 18.06%, 13.06%

7. A firm has net income of $474 million, total equity of $2,625 million and 33 million shares outstanding. The market price is $115 per share. What are its P/E ratio and market-to-book ratio?
a. P/E = 14.36; M/B = 79.55
b. P/E = 8.01; M/B = 1.45
c. P/E = 1.45; M/B = 8.01
d. P/E = 8.01; M/B = 0.69

8. A firm has 33 million shares at $115 each, total liabilities (book value) of $1,005 million, cash of $98 million, EBIT of $741 million and depreciation of $276 million. What is its enterprise value / EBITDA ratio?
a. 4.72
b. 4.62
c. 3.73
d. 6.35

9. Northwind: sales $5,000M, net income $1,050M, total assets $5,000M and total equity $2,500M. Using the DuPont identity, what is its ROE?
a. 21.0%
b. 10.5%
c. 31.5%
d. 42.0%

Chapter 3: Working with Financial Statements — Answers

1. c
Common-size balance sheet accounts are divided by total assets: 84,600 / 470,000 = 18.0%.

2. b
Quick ratio = (current assets − inventory) / current liabilities = (708 − 422) / 548 = 0.52.

3. d
Total debt = 3,630 − 2,625 = 1,005; D/E = 1,005 / 2,625 = 0.38; EM = TA / TE = 3,630 / 2,625 = 1.38 = 1 + D/E.

4. d
TIE = EBIT / interest = 741 / 141 = 5.26; cash coverage = (EBIT + depreciation) / interest = 1017 / 141 = 7.21.

5. a
Inventory turnover = COGS / inventory = 1,344 / 422 = 3.18 times; a higher turnover generally indicates better performance.

6. c
Profit margin = NI / sales = 474 / 2,361 = 20.08%; ROA = NI / TA = 474 / 3,630 = 13.06%; ROE = NI / TE = 474 / 2,625 = 18.06%.

7. b
EPS = 474 / 33 = 14.36, so P/E = 115 / 14.36 = 8.01; book value per share = 2,625 / 33 = 79.55, so M/B = 115 / 79.55 = 1.45.

8. b
EV = 3,795 + 1,005 − 98 = 4,702; EBITDA = 741 + 276 = 1,017; EV/EBITDA = 4,702 / 1,017 = 4.62.

9. d
PM = 1,050 / 5,000 = 21%; TAT = 5,000 / 5,000 = 1.00; EM = 5,000 / 2,500 = 2.00; ROE = PM × TAT × EM = 0.21 × 1.00 × 2.00 = 42.0%.

Chapter 4: Introduction to Valuation — Time Value of Money

Multiple Choice

1. You invest $1,000 for 3 years at 10% per year. How much more do you earn with compound interest than with simple interest?
a. $31
b. $331
c. $100
d. $0

2. You invest $2,000 today at 8% per year, compounded annually. How much will you have after 5 years?
a. $1,361.17
b. $2,800.00
c. $2,938.66
d. $2,720.98

3. You want $10,000 in 6 years and can earn 7% per year, compounded annually. How much must you invest today?
a. $7,042.25
b. $15,007.30
c. $6,663.42
d. $9,345.79

4. An investment of $5,000 grows to $10,000 in 9 years. What annual rate did it earn?
a. 7.18%
b. 8.01%
c. 9.05%
d. 11.11%

5. How long does it take for $1,000 to grow to $2,500 at 9% per year, compounded annually?
a. 10.18 years
b. 16.67 years
c. 11.63 years
d. 10.63 years

6. An investment of $18,000 today will be worth $31,000 in 7 years. Assuming P/Y = 1 and cleared TVM registers, which BA II Plus sequence returns the annual rate of return?

a. 7 N → 18,000 +/− PV → 31,000 FV → 0 PMT → CPT I/Y

b. 7 N → 31,000 +/− PV → 18,000 FV → 0 PMT → CPT I/Y

c. 7 N → 18,000 PV → 31,000 FV → 0 PMT → CPT I/Y

d. 7 N → 18,000 +/− PV → 31,000 FV → 0 PMT → CPT N

Chapter 4: Introduction to Valuation — Time Value of Money — Answers

1. a
Simple: 1,000 + 3 × 100 = 1,300; compound: 1,000 × $(1.10)^3$ = 1,331; the extra $31 is interest on interest.

2. c
FV = 2,000 × $(1.08)^5$ = $2,938.66. BA II Plus: 2nd [CLR TVM], 5 N, 8 I/Y, 0 PMT, 2000 +/− PV, CPT FV.

3. c
PV = 10,000 / $(1.07)^6$ = $6,663.42. BA II Plus: 6 N, 7 I/Y, 0 PMT, 10000 FV, CPT PV gives −6,663.42 (the cash you pay in).

4. b
r = $(10{,}000 / 5{,}000)^{1/9} − 1$ = 8.01%. BA II Plus: 9 N, −5000 PV, 0 PMT, 10000 FV, CPT I/Y.

5. d
2,500 = 1,000 × $(1.09)^t$, so t = ln(2.5) / ln(1.09) = 10.63 years. BA II Plus: 9 I/Y, −1000 PV, 0 PMT, 2500 FV, CPT N.

6. a
Cash paid out is negative and cash received is positive, so PV = −18,000 and FV = +31,000; with N = 7 and PMT = 0, CPT I/Y returns 8.08 (% per year).

Chapter 5: Discounted Cash Flow Valuation

Multiple Choice

1. You deposit $4,000 today and another $7,000 at the end of year 1. The account earns 8% per year. How much is in the account at the end of year 3?
a. $13,203.65
b. $11,000.00
c. $13,856.83
d. $12,225.60

2. An investment pays $200, $400, $600 and $800 at the end of years 1 through 4. The required return is 12%. What is it worth today?
a. $1,604.88
b. $1,271.04
c. $1,432.93
d. $2,000.00

3. You can afford a car payment of $632 per month for 48 months, and the bank quotes 1% per month. How large a loan can you take?
a. $24,239.54
b. $23,999.54
c. $2,466.04
d. $30,336.00

4. You finance $15,000 of a car over 4 years at 9% APR compounded monthly. What is the monthly payment?
a. $1,371.92
b. $385.84
c. $312.50
d. $373.28

5. An investment costs $1,000 today and pays $200 at the end of each year for 8 years. What annual return does it offer?
a. 20.00%
b. 7.50%
c. 6.05%
d. 11.81%

6. You deposit $400 at the end of every month into an account earning 0.5% per month. How many years until the balance reaches $30,000?
a. 5.32 years
b. 63.85 years
c. 75.00 years
d. 6.25 years

7. You put $2,500 into an index fund at the end of every year for 25 years, earning 7% per year. What is the balance at the end?
a. $169,191.18
b. $62,500.00
c. $158,122.59
d. $13,568.58

8. A two-year apartment lease requires $1,800 at the beginning of every month. The relevant rate is 0.5% per month, compounded monthly. What is the lease worth today?
a. $39,016.22
b. $40,816.22
c. $43,200.00
d. $40,613.16

9. A company has just paid a dividend of $2.00 per share. Dividends are expected to grow 3% per year forever, and investors require 10% per year. What is the share worth?
a. $20.00
b. $29.43
c. $28.57
d. $68.67

10. Bank A quotes 15% APR compounded daily; Bank B quotes 15.5% APR compounded annually. What are their EARs, and where would you rather deposit?
a. A = 16.18%; B = 16.76%; Bank B
b. A = 15.56%; B = 15.50%; Bank A
c. A = 15.00%; B = 15.50%; Bank B
d. A = 16.18%; B = 15.50%; Bank A

11. You borrow $10,000 for 4 years at 8%, repaid in equal annual payments of $3,019.21. What is the loan balance after two years?
a. $5,384.05
b. $7,780.79
c. $5,561.58
d. $3,961.58

Chapter 5: Discounted Cash Flow Valuation — Answers

1. a
FV = 4,000 × $(1.08)^3$ + 7,000 × $(1.08)^2$ = 5,038.85 + 8,164.80 = $13,203.65; the $4,000 is invested 3 years and the $7,000 only 2.

2. c
PV = 200/1.12 + 400/$(1.12)^2$ + 600/$(1.12)^3$ + 800/$(1.12)^4$ = $1,432.93; each cash flow is discounted by its own t and then added.

3. b
PV = 632 × [1 − 1/$(1.01)^{48}$] / 0.01 = $23,999.54. BA II Plus: 48 N, 1 I/Y, −632 PMT, 0 FV, CPT PV; N counts payments, not years.

4. d
r = 9% / 12 = 0.75% per month and t = 48: 15,000 = C × [1 − 1/$(1.0075)^{48}$] / 0.0075, so C = $373.28. BA II Plus: 48 N, 0.75 I/Y, 15000 PV, 0 FV, CPT PMT.

5. d
1,000 = 200 × [1 − 1/$(1 + r)^8$] / r cannot be solved by algebra; the calculator iterates: 8 N, −1000 PV, 200 PMT, 0 FV, CPT I/Y = 11.81%.

6. a
30,000 = 400 × [$(1.005)^t$ − 1] / 0.005 gives t = 63.85 months (CPT N); 63.85 / 12 = 5.32 years.

7. c
FV = 2,500 × [$(1.07)^{25}$ − 1] / 0.07 = $158,122.59. BA II Plus: 25 N, 7 I/Y, −2500 PMT, 0 PV, CPT FV.

8. b
PV = ordinary annuity × (1 + r) = 1,800 × [1 − 1/$(1.005)^{24}$] / 0.005 × 1.005 = 40,613.16 × 1.005 = $40,816.22.

9. b
C_1 is the cash flow one period from today: D_1 = 2.00 × 1.03 = 2.06; PV = D_1 / (r − g) = 2.06 / 0.07 = $29.43.

10. d
EAR(A) = $(1 + 0.15/365)^{365}$ − 1 = 16.18%; EAR(B) = 15.50% because compounding is annual; Bank A pays more. BA II Plus ICONV: NOM = 15, C/Y = 365, CPT EFF.

11. a
Balance = PV of the remaining 2 payments = 3,019.21 × [1 − 1/$(1.08)^2$] / 0.08 = $5,384.05.

Chapter 6: Interest Rates and Bond Valuation

Multiple Choice

1. A bond has a $1,000 face value, an 8% annual coupon and 10 years to maturity. Comparable bonds yield 10%. What is its price?
a. $1,000.00
b. $385.54
c. $877.11
d. $1,122.89

2. A $1,000 bond pays a 9% coupon semiannually and matures in 7 years. Comparable bonds yield 8% (quoted annual rate). What is its price?
a. $1,528.16
b. $1,000.00
c. $1,052.82
d. $1,052.06

3. A 5-year zero coupon bond with a $1,000 face value is priced to yield 6% compounded semiannually. What is its price?
a. $558.39
b. $744.09
c. $862.61
d. $747.26

4. A $1,000 bond with an 8% annual coupon and 8 years to maturity trades at $918.50. What is its yield to maturity?
a. 8.7%
b. 10.0%
c. 8.0%
d. 9.5%

5. A $1,000 bond with an 8% annual coupon and 8 years to maturity is priced at $918.50. What is its current yield?
a. 8.71%
b. 9.50%
c. 8.87%
d. 8.00%

6. A $1,000 bond pays an 8% coupon semiannually ($40 per coupon). Two months have passed since the last coupon, and the bond is quoted at $985. What is the invoice (dirty) price?

a. $998.33

b. $1,025.00

c. $985.00

d. $991.67

7. You invest $1,000 for one year at a nominal rate of 15.5%. Inflation is 5%. What is the real rate?

a. 16.28%

b. 10.50%

c. 10.00%

d. 14.76%

8. An investor in the 35% tax bracket compares a 6% municipal bond with a taxable bond. What is the equivalent taxable yield of the muni?

a. 7.50%

b. 9.23%

c. 6.00%

d. 3.90%

Chapter 6: Interest Rates and Bond Valuation — Answers

1. c
Price = $80 \times [1 - 1/(1.10)^{10}] / 0.10 + 1{,}000 / (1.10)^{10} = \$877.11$. BA II Plus: 10 N, 10 I/Y, 80 PMT, 1000 FV, CPT PV = −877.11.

2. c
Work in periods: N = 14, I/Y = 8% / 2 = 4%, PMT = 90 / 2 = 45, FV = 1,000; price = $1,052.82.

3. b
A zero pays only at maturity: price = $1{,}000 / (1.03)^{10} = \$744.09$ (N = 10, I/Y = 3).

4. d
$918.50 = 80 \times [1 - 1/(1 + r)^{8}] / r + 1{,}000 / (1 + r)^{8}$ gives r = 9.5% (9% prices at $944.65 and 10% at $893.30). BA II Plus: 8 N, −918.50 PV, 80 PMT, 1000 FV, CPT I/Y.

5. a
Current yield = annual coupon / market price = 80 / 918.50 = 8.71%; it counts the coupon but ignores the capital gain at maturity.

6. a
Accrued interest = 40 × (2 months / 6 months) = 13.33; dirty price = clean price + accrued interest = 985 + 13.33 = $998.33.

7. c
Fisher: $1 + R = (1 + r)(1 + h)$, so $r = 1.155 / 1.05 - 1 = 10.00\%$.

8. b
Equivalent taxable yield = muni yield / (1 − t) = 6% / 0.65 = 9.23%.

Chapter 7: Equity Markets and Stock Valuation

Multiple Choice

1. A company just paid a dividend of $2.00 per share. Dividends are expected to grow at 5% a year indefinitely, and investors require a 12% return. What is the stock worth?
a. $17.50
b. $28.57
c. $42.00
d. $30.00

2. A company just paid a dividend of $2.00 per share. Dividends are expected to grow at 5% a year indefinitely, and investors require a 12% return. What is the expected price five years from today, just after the fifth dividend has been paid?
a. $36.47
b. $30.00
c. $40.20
d. $38.29

3. A firm just paid a $1.00 dividend. Dividends will grow at 20% for the next three years and at 5% forever after that. The required return is 15%. What is the stock worth today?
a. $15.20
b. $14.63
c. $13.64
d. $3.27

4. A firm will pay no dividend for the next four years. It will pay $0.50 at the end of year 5, and that dividend will then grow at 4% forever. The required return is 20%. What is the stock worth today?
a. $3.12
b. $1.21
c. $1.51
d. $1.26

5. A stock sells for $20.00. Its next dividend will be $1.00, and dividends are expected to grow at 10% a year. What are its dividend yield, capital gains yield and required return?
a. 5.0%; 10.0%; 5.0%
b. 5.0%; 10.0%; 15.0%
c. 5.0%; 5.0%; 10.0%
d. 10.0%; 5.0%; 15.0%

6. A firm is expected to earn $3.00 per share next year. Comparable firms trade at 18 times earnings. What is the estimated share price?

a. $18.00

b. $54.00

c. $0.17

d. $21.00

7. A loss-making firm has sales of $10.00 per share. Comparable firms trade at 2.5 times sales. What is the estimated share price?

a. $4.00

b. $12.50

c. $0.00

d. $25.00

Chapter 7: Equity Markets and Stock Valuation — Answers

1. d
$D_1 = 2.00 \times 1.05 = 2.10$; $P_0 = D_1 / (R - g) = 2.10 / (0.12 - 0.05) = \$30.00$.

2. d
The model works from any date if you use the following period's dividend: $P_5 = D_6 / (R - g) = 2.00 \times (1.05)^6 / 0.07 = \$38.29$.

3. a
$D_1 = 1.20$, $D_2 = 1.44$, $D_3 = 1.728$, $D_4 = 1.728 \times 1.05 = 1.8144$; $P_3 = D_4 / (0.15 - 0.05) = 18.144$; PV $= 1.20/1.15 + 1.44/(1.15)^2 + 1.728/(1.15)^3 + 18.144/(1.15)^3 = \$15.20$.

4. c
Apply the model one period before the first dividend: $P_4 = 0.50 / (0.20 - 0.04) = 3.125$; $P_0 = 3.125 / (1.20)^4 = \$1.51$.

5. b
Dividend yield $= D_1 / P_0 = 1.00 / 20.00 = 5.0\%$; under constant growth the price grows at g, so the capital gains yield $= g = 10.0\%$; $R = 5.0\% + 10.0\% = 15.0\%$.

6. b
Price = EPS × benchmark P/E $= 3.00 \times 18 = \$54.00$; the answer is only as good as the word “comparable”.

7. d
Price = sales per share × benchmark price-sales ratio $= 10.00 \times 2.5 = \$25.00$; sales are always positive, so P/S still works when earnings are negative and a P/E ratio is meaningless.